Cambridge Assessment International Education

Cambridge O Level

BIOLOGY **5090/31**

Paper 3 Practical Test **May/June 2023**

CONFIDENTIAL INSTRUCTIONS

L1490646 48

This document gives details of how to prepare for and administer the practical exam.

The information in this document and the identity of any materials supplied by Cambridge International are confidential and must NOT reach candidates either directly or indirectly.

The supervisor must complete the report at the end of this document and return it with the scripts.

INSTRUCTIONS

- If you have any queries regarding these confidential instructions, contact Cambridge International stating the centre number, the syllabus and component number and the nature of the query.
  email info@cambridgeinternational.org
  phone +44 1223 553554

This document has **8** pages. Any blank pages are indicated.

DC (DE) 326137
© UCLES 2023

General information about practical exams

Centres must follow the guidance on science practical exams given in the *Cambridge Handbook*.

Safety

Supervisors must follow national and local regulations relating to safety and first aid.

Only those procedures described in the question paper should be attempted.

Supervisors must inform candidates that materials and apparatus used in the exam should be treated with caution. Suitable eye protection should be used where necessary.

The following hazard codes are used in these confidential instructions, where relevant:

C	corrosive	**MH**	moderate hazard
HH	health hazard	**T**	acutely toxic
F	flammable	**O**	oxidising
N	hazardous to the aquatic environment		

Hazard data sheets relating to substances used in this exam should be available from your chemical supplier.

Before the exam

- The packets containing the question papers must **not** be opened before the exam.
- It is assumed that standard school laboratory facilities, as indicated in the *Guide to Planning Practical Science*, will be available.
- Spare materials and apparatus for the tasks set must be available for candidates, if required.

During the exam

- It must be made clear to candidates at the start of the exam that they may request spare materials and apparatus for the tasks set.
- Where specified, the supervisor **must** perform the experiments and record the results as instructed. This must be done **out of sight** of the candidates, using the same materials and apparatus as the candidates.
- Any assistance provided to candidates must be recorded in the supervisor's report.
- If any materials or apparatus need to be replaced, for example, in the event of breakage or loss, this must be recorded in the supervisor's report.

After the exam

- The supervisor must complete a report for each practical session held and each laboratory used.
- Each packet of scripts returned to Cambridge International must contain the following items:
  - the scripts of the candidates specified on the bar code label provided
  - the supervisor's results relevant to these candidates
  - the supervisor's reports relevant to these candidates
  - seating plans for each practical session, referring to each candidate by candidate number
  - the attendance register.

© UCLES 2023

Specific information for this practical exam

During the exam, the supervisor (**not** the invigilator) must do the experiment in Question 1 and record the results on a spare copy of the question paper, clearly labelled 'supervisor's results'.

Apparatus and materials

Question 1

Each student will require:

(i) four cylinders of uncooked Irish potato with outer skin removed. Each cylinder should be approximately 50 mm in length (see Note 1).

(ii) white tile

(iii) scalpel or sharp knife

(iv) 15 cm^3 distilled or deionised water labelled **0%**

(v) 15 cm^3 of a 2.0% sodium chloride solution labelled **2.0%** (see Note 2)

(vi) 15 cm^3 of a 4.0% sodium chloride solution labelled **4.0%**

(vii) 15 cm^3 of a 6.0% sodium chloride solution labelled **6.0%**

(viii) four test-tubes and test-tube rack or similar

(ix) means of writing on glassware (e.g. wax pencil or water-resistant marker)

(x) beaker approximately 250 cm^3 labelled **waste**

(xi) stop-watch, timer or sight of clock

(xii) 30 cm ruler graduated in millimetres

(xiii) paper towels.

Note 1
Potato cylinders should be freshly cut using a cork borer with a diameter of approximately 6 mm. They should be placed in a Petri dish or similar and covered with a damp paper towel.

Note 2
To make up a 10.0% stock solution of sodium chloride, dissolve 100 g sodium chloride in 1 dm^3 distilled water.

To make a 2.0% solution of sodium chloride, add 2 parts by volume of this stock solution to 8 parts distilled water.

To make a 4.0% solution of sodium chloride, add 4 parts by volume of this stock solution to 6 parts distilled water.

To make a 6.0% solution of sodium chloride, add 6 parts by volume of this stock solution to 4 parts distilled water.

© UCLES 2023

Question 2

(i) 30 cm ruler graduated in millimetres

© UCLES 2023

BLANK PAGE

© UCLES 2023

[Turn over

BLANK PAGE

Permission to reproduce items where third-party owned material protected by copyright is included has been sought and cleared where possible. Every reasonable effort has been made by the publisher (UCLES) to trace copyright holders, but if any items requiring clearance have unwittingly been included, the publisher will be pleased to make amends at the earliest possible opportunity.

To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced online in the Cambridge Assessment International Education Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download at www.cambridgeinternational.org after the live examination series.

Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

© UCLES 2023

Supervisor's report

Syllabus and component number | | | | | / | | |

Centre number | | | | | |

Centre name .....

Time of the practical session .....

Laboratory name/number .....

Give details of any difficulties experienced by the centre or by candidates (include the relevant candidate names and candidate numbers).

You must include:

- any difficulties experienced by the centre in the preparation of materials
- any difficulties experienced by candidates, e.g. due to faulty materials or apparatus
- any specific assistance given to candidates.

© UCLES 2023

[Turn over

Declaration

1 Each packet that I am returning to Cambridge International contains all of the following items:
- the scripts of the candidates specified on the bar code label provided
- the supervisor's results relevant to these candidates
- the supervisor's reports relevant to these candidates
- seating plans for each practical session, referring to each candidate by candidate number
- the attendance register.

2 Where the practical exam has taken place in more than one practical session, I have clearly labelled the supervisor's results, supervisor's reports and seating plans with the time and laboratory name/number for each practical session.

3 I have included details of difficulties relating to each practical session experienced by the centre or by candidates.

4 I have reported any other adverse circumstances affecting candidates, e.g. illness, bereavement or temporary injury, directly to Cambridge International on a *special consideration form*.

Signed ..... (supervisor)

Name (in block capitals) .....

© UCLES 2023